Exp: 4 Date:

Implement Packet Sniffing Using raw sockets in python.

Aim: To implement pocket sniffing using raw sockets in python & capture network packets passing through network interface

objective: To understand how raw sockets work & how network packets can be captured & analyze using python programming.

Theory:

1. Pocket Sniffing is the process of monitoring & capturing data packets travelling across a network

2. A pocket sniffer intercepts packets & analyzes their contents

3. Raw Sockets provide direct access to Lower-level network protocols, allowing applications to capture Packets

4. In python, the socket Module, can be used to create raw sockets for packet capture.

5. This Technique is useful for networking monitoring trouble shooting & security Analysis

Algorithm:

1. Import the required socket Module
2. Creat a raw socket
3. Bind the socket to the networks interface
4. Receive Packets using the recvfrom()
5. display the captured packet information
6. Repeat the process continuosly to monitor network traffic.

Program:

```
import socket:
    Sniffer = socket.Socket(socket.
    AF_INET, SOCKET, SOCK_RAW,
  SOCKET.IPROTIO_IP)
Sniffer.bind((host,0))
while True:
    raw-data, addr=Sniffer.recvFrom
                        (65535)
Print("Packet received From:", Addr)
Print(raw_data)
```

Output:-

Packet received from: ('192.168.15', 0)
b'E\x00\x00 c...\

Packet received from: ('192.168.1.10', 0)
b'E\x00\x00c ...

Result: Thus Packet Sniffing was successfully implemented Using raw sockets in python & network Packets were captured and displayed